

Euclidean Geometry In Mathematical Olympiads

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Problem 1.43 (JMO 2011/5)

[Chapter 1](#)[fgm](#)[fgmcon](#)[Bookmark](#)[Reply](#)midpoint of AC .

Solution with directed angles

Because of the tangency criterion we have $\angle PBD = \angle BED = x$. Now since $ED \parallel AC$ we see $\angle EYC = \angle BYA = \angle BED = \angle BOP$. $\angle BYA = \angle BYP$ so $BYOP$ are all concyclic. Now with this we deduce $\angle OBP = \angle OYP = 90^\circ$ so $OY \perp AC$. Combined with $OC = OA$ we deduce that $YC = YA$ as desired.

This post has been edited 1 time. Last edited by Jerry122805, Aug 13, 2020, 7:24 PM

ilovepizza2020

12149 posts

May 9, 2021, 10:33 AM

Just look up 2011 JMO P5 in contest collections lawl

#9

Just got to this problem in the book 😊

serpent_121

1224 posts

Jun 21, 2021, 2:56 AM • 1 🍊

Do we need to direct angles? I don't see any real configuration issues; we can always WLOG A is closer to P than C right?

#10

MrOreoJuice

600 posts

Jun 21, 2021, 3:15 AM

~~You don't even need any angles if you use projective~~

#11

ike.chen

1162 posts

Jun 21, 2021, 4:15 AM



serpent_121 wrote:

Do we need to direct angles? I don't see any real configuration issues; we can always WLOG A is closer to P than C right?

#12

Umm I guess AC could be closer to D than B (and vice versa) so technically there's config issues but no one really cares lol (i.e. I only use directed when it's actually convenient).

samrocksnature

8780 posts

Aug 11, 2021, 1:22 AM • 1 🍊

Take the problem mod B to get that $(\text{mod } B)$ is cyclic.

#13



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